

sdstudio

A Companion Package for Designing
and Managing surveydown Surveys

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About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in sustainable transportation and human behavior
 2. **Open-Source Software** in **R** tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfan.org

jhelvy.shinyapps.io

 **Welcome to the Ultimate Superhero Survey!** 

Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

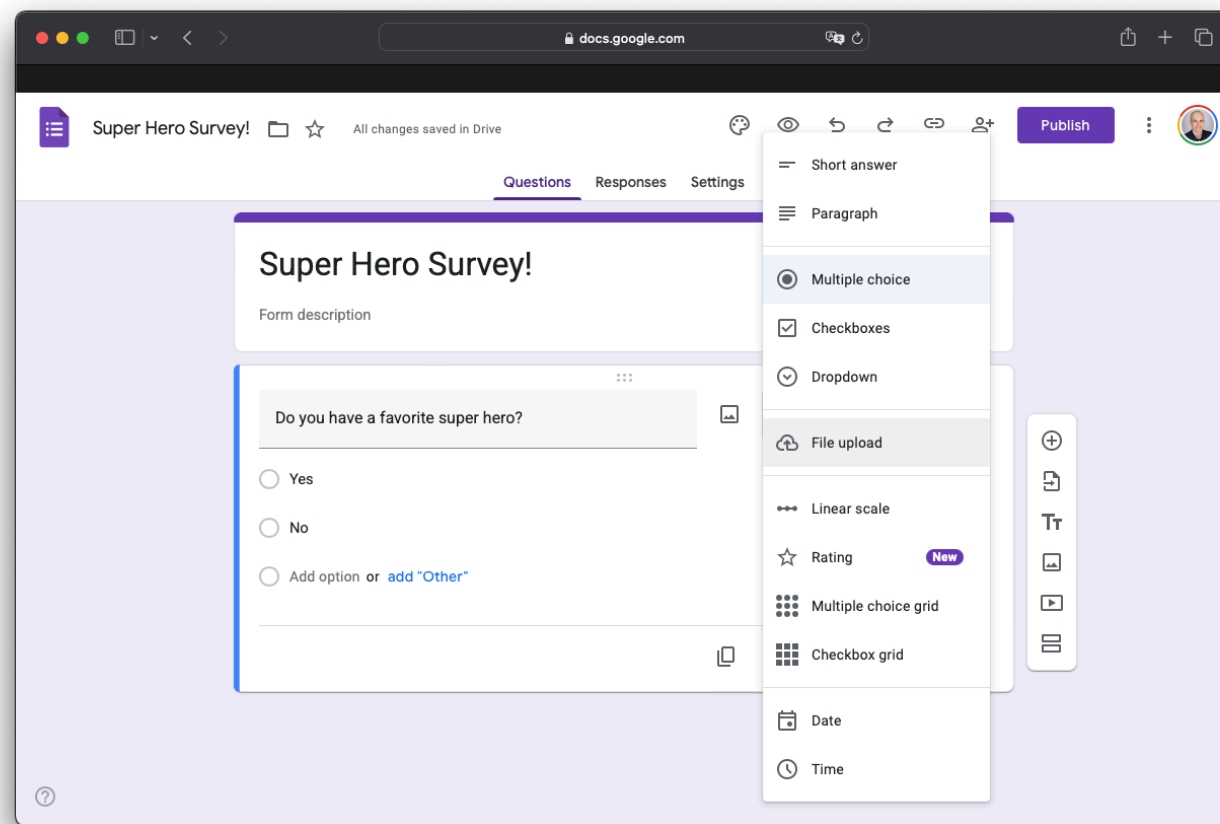
Do you have a favorite super hero?*

☐ Yes

☒ No

Next

WYSIWYG Interface



- ⊗ Reproducibility
- ⊗ Version control
- ⊗ Limited features
- ⊗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



Survey design using Quarto and Shiny



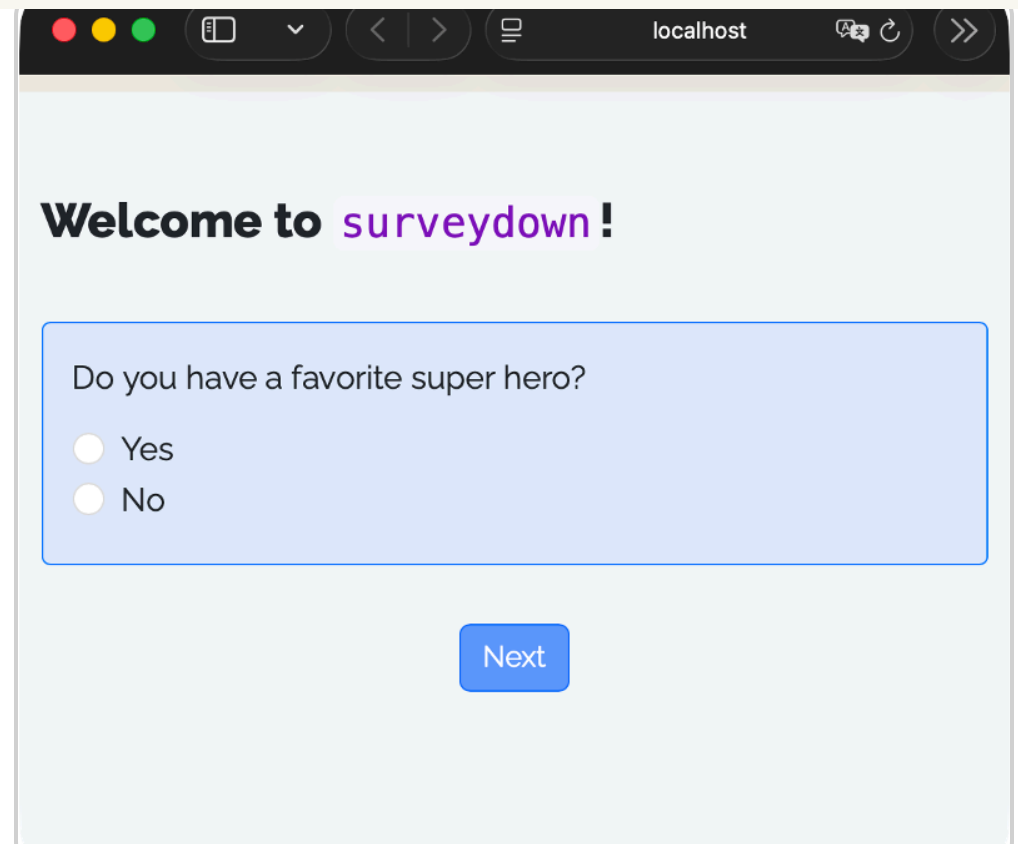
+



+



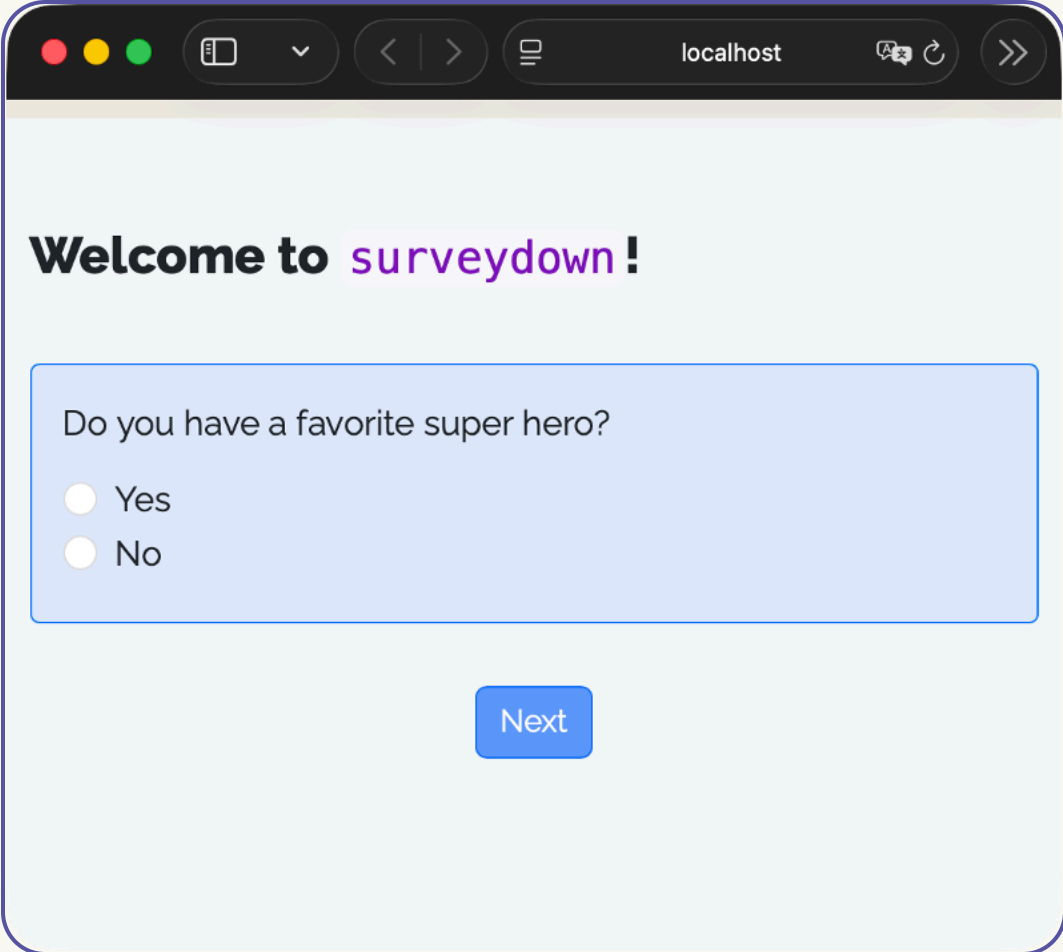
=



survey.qmd

```
1 ---
2 survey-settings:
3   use-cookies: true
4   rate-survey: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite super hero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25 ```
26
27 --- some_other_page
28
29 Other content...
```

Rendered Survey



localhost

Welcome to surveydown!

Do you have a favorite super hero?

☐ Yes

☐ No

Next

But...



Code-Only Interface

No GUI. Not everyone is comfortable working purely in code.



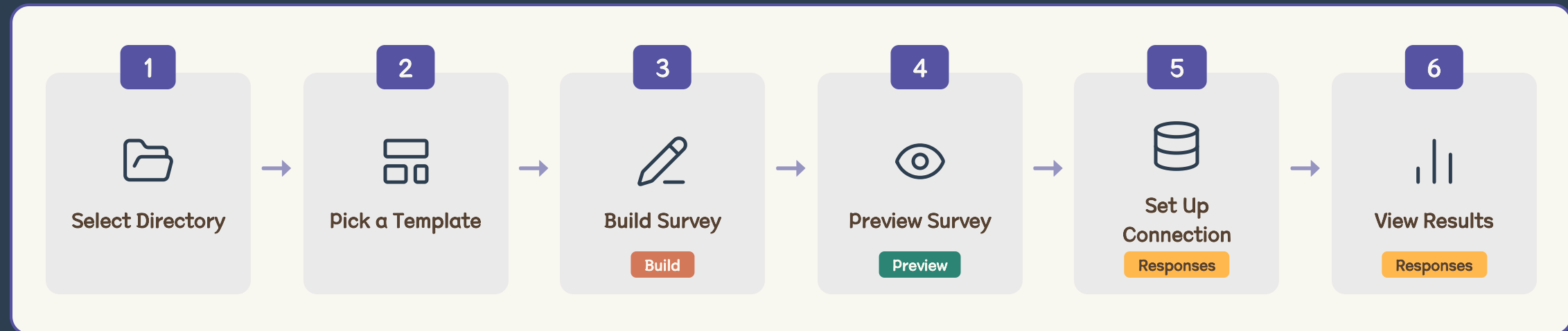
Learning Curve

The survey pipeline is not obvious. It takes time to learn.

Introducing **sdstudio**!



Studio Workflow



Launch with a simple command

```
sdstudio :: launch()
```

The image shows a web browser window with the address bar displaying '127.0.0.1'. The page title is 'surveydown Studio'. The navigation bar includes 'Build' (underlined), 'Preview', and 'Responses'. In the top right corner, there are icons for a folder, a plus sign, and a share icon. The main content area features a large heading 'Create New Survey'. Below this, there are two form fields: 'Template:' with a dropdown menu showing 'Default', and 'Directory:' with a text input field containing 'survey/'. At the bottom center, there is a blue button labeled 'Create Survey'.

Start with a Template

The screenshot shows a web browser window with the address bar displaying '127.0.0.1'. The page title is 'surveydown Studio'. The navigation bar includes 'Build' (active), 'Preview', and 'Responses'. In the top right corner, there are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey'. It contains two form fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

surveydown.org/templates

surveydown.org

About

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1 surveydown Studio

Templates – surveydown.org

Templates

Basic

Default Template

Question Types

Questions Defined with YAML

Conditions

Conditional Showing

Conditional Skipping

Conditional Stopping

Randomization

Random Options

Random Options Predefined

Option Shuffling

Reactivity

Reactive Questions

Reactive drilldown

External Redirect

Live Polling

Conjoint

Conjoint Buttons

Conjoint Tables

Custom

Custom Leaflet Map

Custom Plotly Chart

Templates

To help you get started with surveydown, we created some templates that cover many of the features of the package. Feel free to use any as a starting point for your own surveys or as a reference for specific features.

Call `sd_create_survey()` with proper arguments to create your desired templates in your preferred directory. Refer to the [Start with a template](#) section for more details.

Default Template

Hi! This is a default template of a surveydown survey. It has two pages with one question on each page.

Here is a basic "multiple choice" question:

Which type of penguin do you like the best?

Adelie

Chinstrap

Gentoo

You need to insert next buttons with `sd_next()` on each page. By default, it will go to the next page defined in the `survey.yml` file, but you can also specify the page with the `next_page` argument set to the name of the page you want to go to next.

Next

Default Template

BASIC

A minimum template for starting from scratch.

Template - Question Types

Hi! This is a showcase of different question types and question formatting that surveydown supports.

Please make your choice below:

Where do you want to go to?

Take me to the page on Question Types (the next page)

Take me to the page on Question Formatting

Skip to the end

Next

Question Types

BASIC

A template showcasing all built-in [question types](#).

Template - Questions Defined with YAML

This is a showcase of defining questions with a `questions.yml` file.

The `sd_question()` allows a `yml` argument default to `"questions.yml"`. If you have such file in your project root folder you can use it to support your question definition.

You can also change to other file name or directory and define questions like this:

```
sd_question(
  id = "your_question_id",
  yml = "folder_name/questions.yml"
)
```

Please click on the "Next" button to see the demo questions.

Next

Questions Defined with YAML

BASIC

A template showcasing all built-in [question types](#) using a YAML file.

Categories

All (17)

Basic (3)

Conditions (3)

Conjoint (2)

Custom (2)

Randomization (3)

Reactivity (4)

Site made with

and

quarto

Choose Your Directory

The screenshot shows a web browser window with the address bar displaying '127.0.0.1'. The browser's top bar includes standard navigation icons (back, forward, refresh) and a settings gear icon. The application's header features the text 'surveydown Studio' followed by three tabs: 'Build' (which is the active tab and underlined), 'Preview', and 'Responses'. On the right side of the header, there are two icons: a yellow square with a white monitor icon and a blue square with a white cloud icon. The main content area is titled 'Create New Survey' in a large, centered font. Below this title, there are two form fields. The first is labeled 'Template:' and contains a dropdown menu with 'Default' selected. The second is labeled 'Directory:' and contains a dark grey input field with the text 'survey/' and a mouse cursor icon at the end. Below these fields is a prominent blue button with the text 'Create Survey' in white.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

The Build Tab

surveydown Studio

Build

Preview

Responses

Structure

Page: welcome

T

Default Template Welcome to...

Q

penguins | mc

What's your favorite penguin?

T

There are several ways to...

Page: page2

T

This is another page in...

Q

silly_word | text

Write a silly word here:

T

On this page, we don't...

Page: end

T

End This is the...

Add A New Page

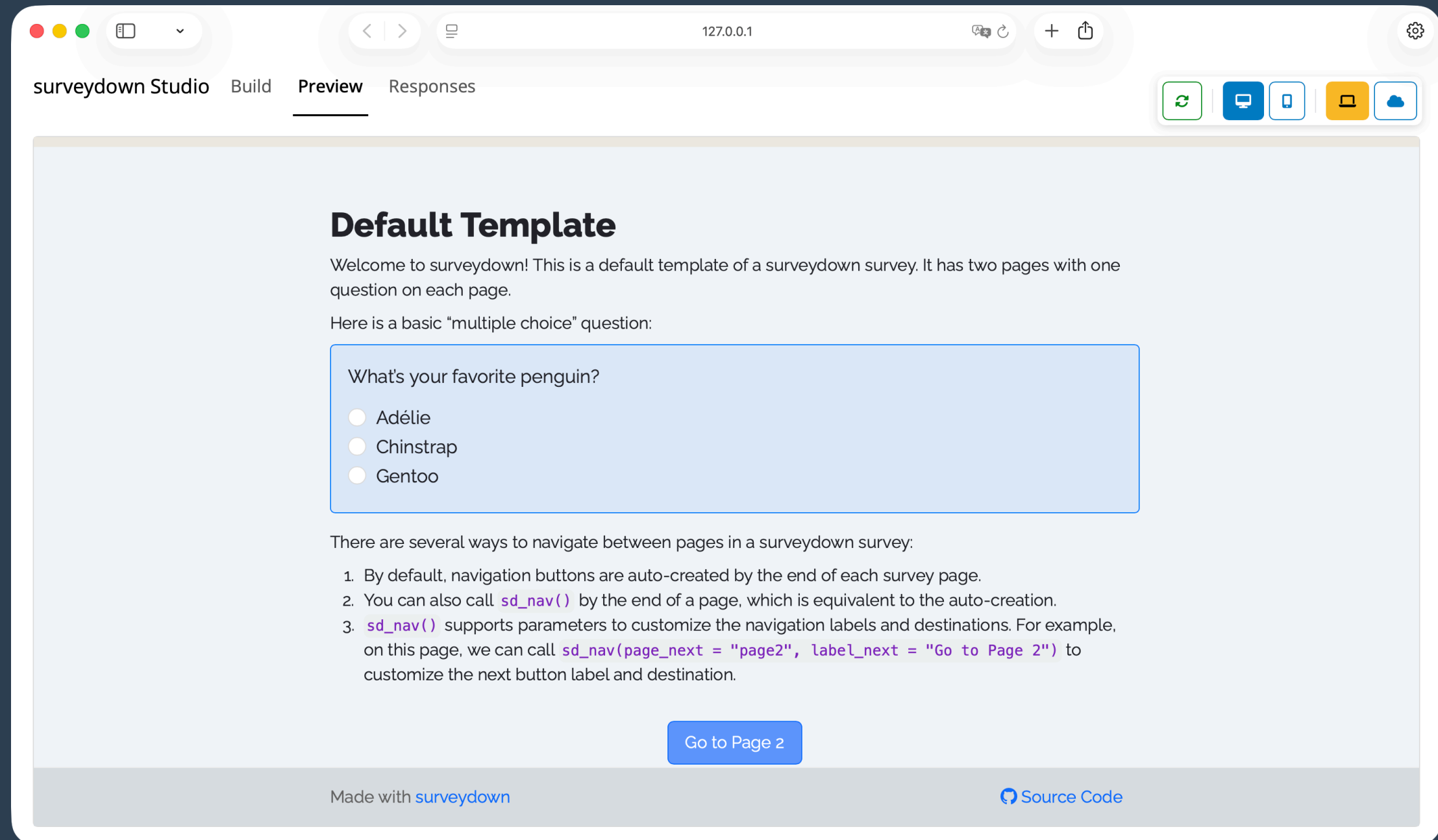
Code

survey.qmd

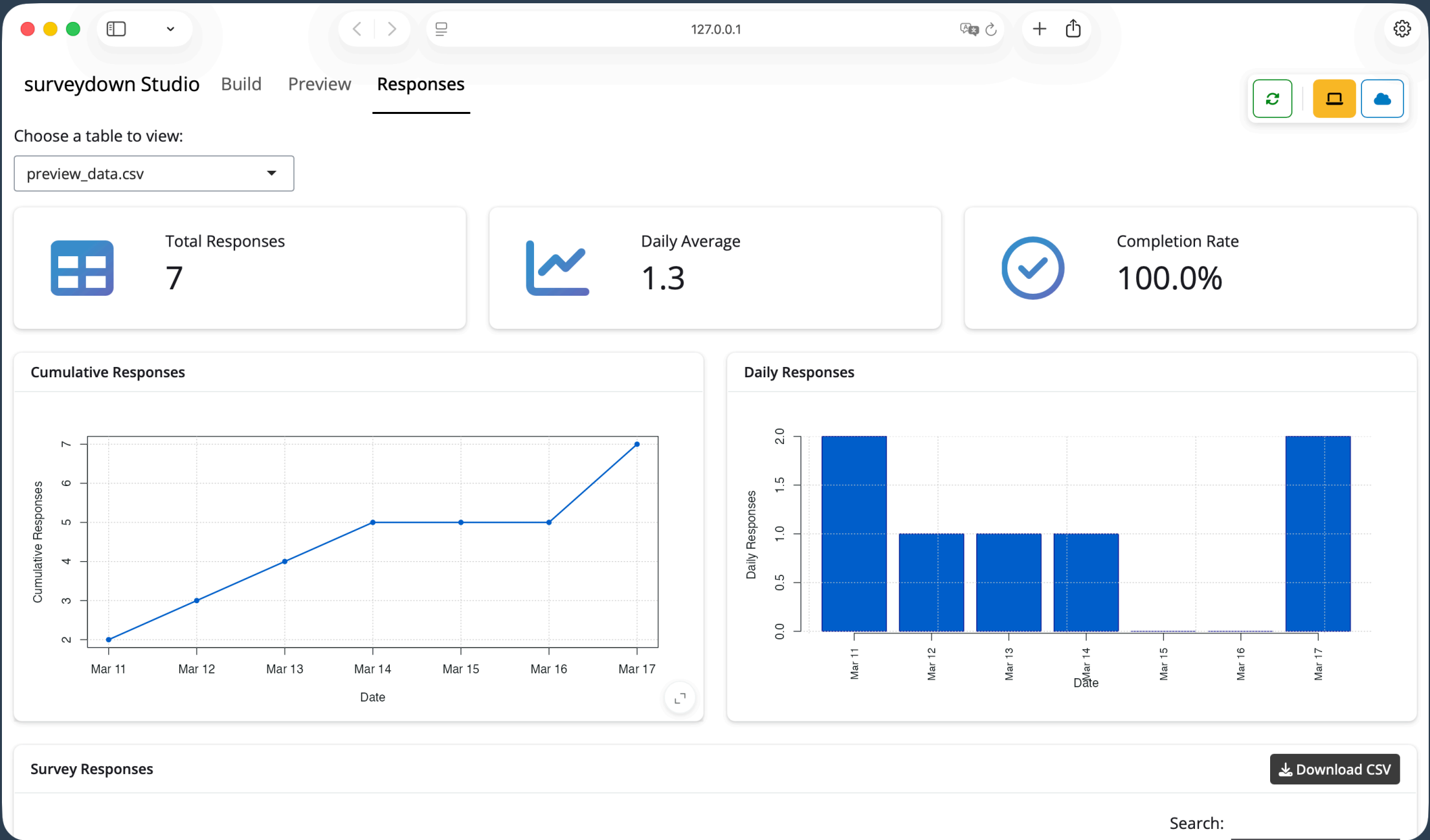
app.R

```
37 --- welcome
38
39 # Default Template
40
41 Welcome to surveydown! This is a default template of a surveydown survey. It has two
42 pages with one question on each page.
43
44 Here is a basic "multiple choice" question:
45
46 ```{r}
47 sd_question(
48   type = 'mc',
49   id = 'penguins',
50   label = "What's your favorite penguin?",
51   option = c(
52     'Adélie' = 'adelie',
53     'Chinstrap' = 'chinstrap',
54     'Gentoo' = 'gentoo'
55   )
56 )
57
58 There are several ways to navigate between pages in a surveydown survey:
59
60 1. By default, navigation buttons are auto-created by the end of each survey page.
61 2. You can also call `sd_nav()` by the end of a page, which is equivalent to the
62    auto-creation.
63 3. `sd_nav()` supports parameters to customize the navigation labels and destinations.
64    For example, on this page, we can call `sd_nav(page_next = "page2", label_next = "Go to
65    Page 2")` to customize the next button label and destination.
66
67 ```{r}
```


The Preview Tab



The Responses Tab



127.0.0.1

Cumulative Responses

Date	Cumulative Responses
Mar 11	2
Mar 12	3
Mar 13	4
Mar 14	5
Mar 15	5
Mar 16	5
Mar 17	~6.5

Daily Responses

Date	Daily Responses
Mar 11	~1.8
Mar 12	~1.0
Mar 13	~1.0
Mar 14	~1.0
Mar 15	~0.0
Mar 16	~0.0
Mar 17	~1.8

Survey Responses

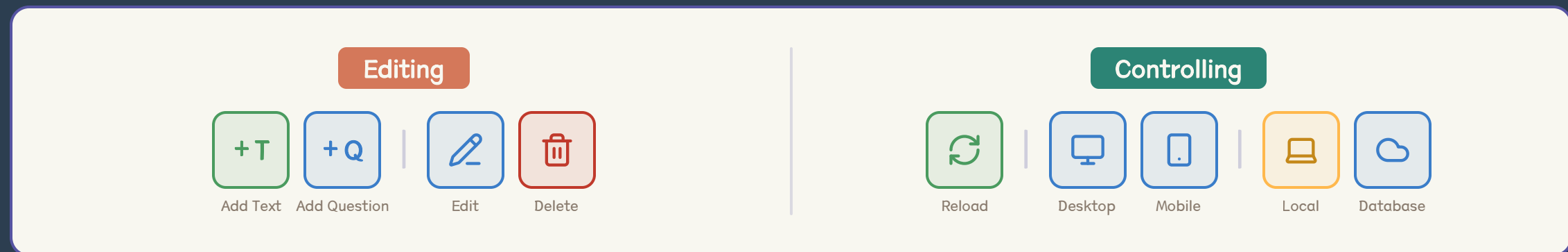
Download CSV

Search:

session_id	time_start	browser	ip_address	time_end	penguins	time_q_penguins	time.
1 b297324fa59944f94cee795a4970bbf9	2026-03-11 20:27:01 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-11 20:27:20 UTC			2026
2 ccf129da18d1efb92edc779acfc5df9f	2026-03-11 20:27:32 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-11 20:34:36 UTC	adelie	2026-03-11 20:28:05 UTC	
3 197755610bc125cdb432282f750e70e4	2026-03-12 20:34:36 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-12 20:34:46 UTC	chinstrap	2026-03-12 20:34:38 UTC	
4 70d2cb7baee64e9770ec02c77a59b9c9	2026-03-13 20:34:46 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-13 20:45:13 UTC	gentoo	2026-03-13 20:34:48 UTC	
5 b91c98cb4b902b235fb0e6df96248800	2026-03-14 22:05:19 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-14 04:01:06 UTC			
6 cf68ddaa8ac748c042224f4c8e127449	2026-03-17 01:58:26 UTC	Safari v26, Mac OS X	127.0.0.1	2026-03-21 02:00:01 UTC	adelie	2026-03-17 01:58:27 UTC	
7 e87acb28aac61b13d6f097f77e280954	2026-03-17 01:58:54 UTC	Electron v39, Mac OS X	127.0.0.1	2026-03-21 01:59:59 UTC			

Studio Features

Self-explanatory Icons



Editing

GUI



Point & Click

No coding required

or

Script



Edit Directly

Full code control

GUI - Toggling

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs, there is a "Structure" section with a refresh icon and a dropdown arrow.
- The structure is organized into pages:
 - Page: page1** (with icons for adding text, adding questions, editing, deleting, and a dropdown):
 - A text element: `# Title Some texts.`
 - A question element: `penguins | mc` with the label `What's your favorite penguin?`
 - Page: end** (with similar icons):
 - A text element: `## End This is the...`
- At the bottom of the Structure panel is an orange button labeled "Add A New Page".

Code Panel:

- The Code panel is titled "Code" and shows the R code for the survey.
- The code is organized into sections corresponding to the Structure panel:
 - `library(surveydown)`
 - `---`
 - `--- page1`
 - `# Title`
 - `Some texts.`
 - ````{r}`
 - `sd_question(`
 - `type = 'mc',`
 - `id = 'penguins',`
 - `label = "What's your favorite penguin?",`
 - `option = c(`
 - `'Adélie' = 'adelie',`
 - `'Chinstrap' = 'chinstrap',`
 - `'Gentoo' = 'gentoo'`
 - `)`
 - `)`
 - `````
 - `---`
 - `end`
 - `## End`
 - `This is the last page in the survey.`
 - ````{r}`
 - `sd_close()`
 - `````

GUI - Content Editing

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs, there is a "Structure" section with a refresh icon and a dropdown arrow.
- The structure is organized into pages:
 - Page: page1** (with icons for adding text, adding questions, editing, and deleting):
 - A text element: `# Title Some texts.`
 - A question element: `penguins | mc` with the label `What's your favorite penguin?`
 - Page: end** (with similar icons):
 - A text element: `## End This is the...`
- At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

- The Code panel shows the R code for the survey, with a file explorer at the top showing `survey.qmd` and `app.R`.
- The code is as follows:

```
33 library(surveydown)
34 
35 
36 
37 --- page1
38 
39 # Title
40 
41 Some texts.
42 
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ```
55 
56 --- end
57 
58 ## End
59 
60 This is the last page in the survey.
61 
62 ```{r}
63 sd_close()
64 ```
```


GUI - Content Addition

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs, there is a "Structure" section with a refresh icon and a "Page: page1" header.
- Under "Page: page1", there are two items:
 - A text item labeled "# Title New texts." with a green edit icon and a red delete icon.
 - A question item labeled "penguins | mc" with the text "What's your favorite penguin?" and a green edit icon and a red delete icon.
- Below "Page: page1", there is a "Page: end" header with a green edit icon and a red delete icon.
- Under "Page: end", there is a text item labeled "## End This is the..." with a green edit icon and a red delete icon.
- At the bottom of the Structure panel, there is an orange button labeled "Add A New Page".

Code Panel:

- The Code panel is titled "Code" and shows the R code for the survey.
- The code is organized into sections: "survey.qmd" and "app.R".
- The "survey.qmd" section contains the following code:

```
33 library(surveydown)
34 ---
35
36
37 --- page1
38
39 # Title
40
41 New texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 ```
```
- The "app.R" section is currently empty.

GUI - Page Addition

The screenshot displays the surveydown Studio interface. The top navigation bar includes 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Structure' panel on the left shows a hierarchical view of the survey pages. It contains two pages: 'Page: page1' and 'Page: end'. 'Page: page1' has three elements: a title '# Title New texts.', a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?', and a text question 'text_question | text' with the label 'Write something:'. 'Page: end' has a single element: a text block '## End This is the...'. Below the structure panel is an orange button labeled 'Add A New Page'. The 'Code' panel on the right shows the R code for the survey. It includes a 'survey.qmd' file and an 'app.R' file. The R code defines two questions: a multiple-choice question 'penguins' and a text question 'text_question'. The code ends with a '## End' block and a closing statement 'This is the last page in the survey.'.

surveydown Studio Build Preview Responses

Structure

Page: page1

- # Title New texts.
- penguins | mc
What's your favorite penguin?
- text_question | text
Write something:

Page: end

- ## End This is the...

Add A New Page

Code

```
survey.qmd app.R
41 new texts.
42
43- ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54- ```
55- ```{r}
56 sd_question(
57   type = "text",
58   id = "text_question",
59   label = "Write something:"
60 )
61- ```
62
63
64 --- end
65
66- ## End
67
68 This is the last page in the survey.
69
70- ```{r}
71 sd_close()
72- ```
```

GUI - Page Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1** (includes +T, +Q, edit, and delete icons):
 - # Title New texts.
 - penguins | mc
What's your favorite penguin?
 - text_question | text
Write something:
- Page: page2** (includes +T, +Q, edit, and delete icons):
 - This is page 2.
- Page: end** (includes +T, +Q, edit, and delete icons):
 - ## End This is the...

An orange button labeled "Add A New Page" is located at the bottom of the Structure panel.

Code Panel:

The Code panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R". The visible code is as follows:

```
41 new texts.  
42  
43- ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie'   = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo'   = 'gentoo'  
52   )  
53 )  
54- ```  
55- ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61- ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71- ## End  
72
```

GUI - Content Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title New texts.** (Text question)
 - penguins | mc** (Multiple choice question: "What's your favorite penguin?")
 - text_question | text** (Text question: "Write something:")
- Page: page2**
 - This is page 2.** (Text question)
- Page: end**
 - ## End This is the...** (Text question)

An orange button labeled "Add A New Page" is located at the bottom of the Structure panel.

Code Panel:

The Code panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R". The visible code is as follows:

```
41 new texts.  
42  
43- ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie'   = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo'   = 'gentoo'  
52   )  
53 )  
54- ```  
55- ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61- ```  
62  
63  
64 --- page2  
65  
66 This is page 2.  
67  
68  
69 --- end  
70  
71- ## End  
72
```


Script Editing

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - Text question: # Title New texts.
 - Multiple choice question: penguins | mc. Label: What's your favorite penguin?
- Page: page2**
 - Text question: This is page 2.
 - Text question: text_question | text. Label: Write something:
- Page: end**
 - Text question: ## End This is the...

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R script for the survey, with tabs for "survey.qmd" and "app.R". The visible code is as follows:

```
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ---
55 --- page2
56
57 This is page 2.
58
59 ```{r}
60 sd_question(
61   type = "text",
62   id = "text_question",
63   label = "Write something:"
64 )
65 ---
66
67 --- end
68
69 ## End
70
71 This is the last page in the survey.
72
73 ```{r}
74 sd_close()
75 ```
```

Instant Preview

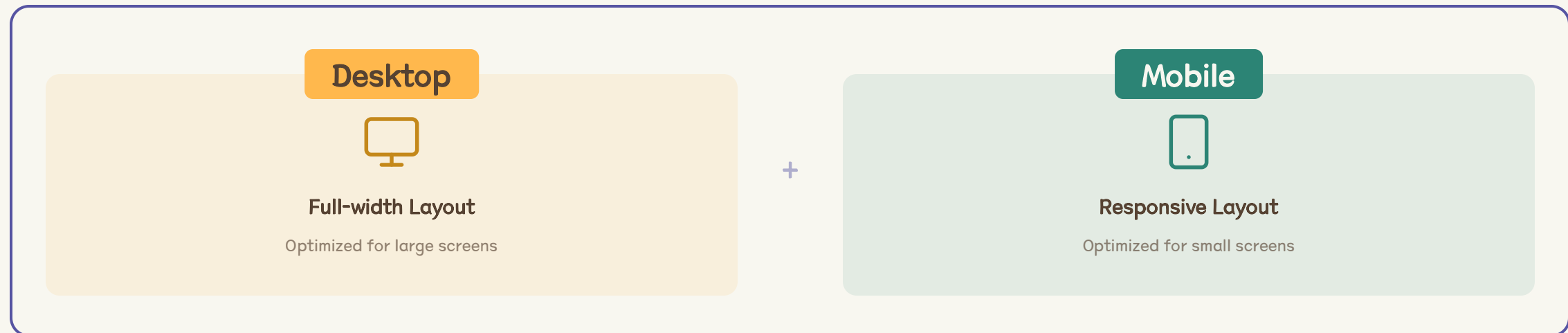
The screenshot displays the surveydown Studio application interface. At the top, there's a navigation bar with tabs for 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Build' tab is currently active. Below the navigation bar, the interface is split into two main panels: 'Structure' on the left and 'Code' on the right.

Structure Panel: This panel shows a hierarchical view of the survey structure. It lists three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page has a set of controls (add text, add question, edit, delete, and expand/collapse). Under 'Page: page1', there is a text block '# Title New texts.' and a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?'. Under 'Page: page2', there is a text block 'This is page 2.' and a text question 'text_question | text' with the label 'Write something:'. At the bottom of the Structure panel, there is an orange button labeled 'Add A New Page'.

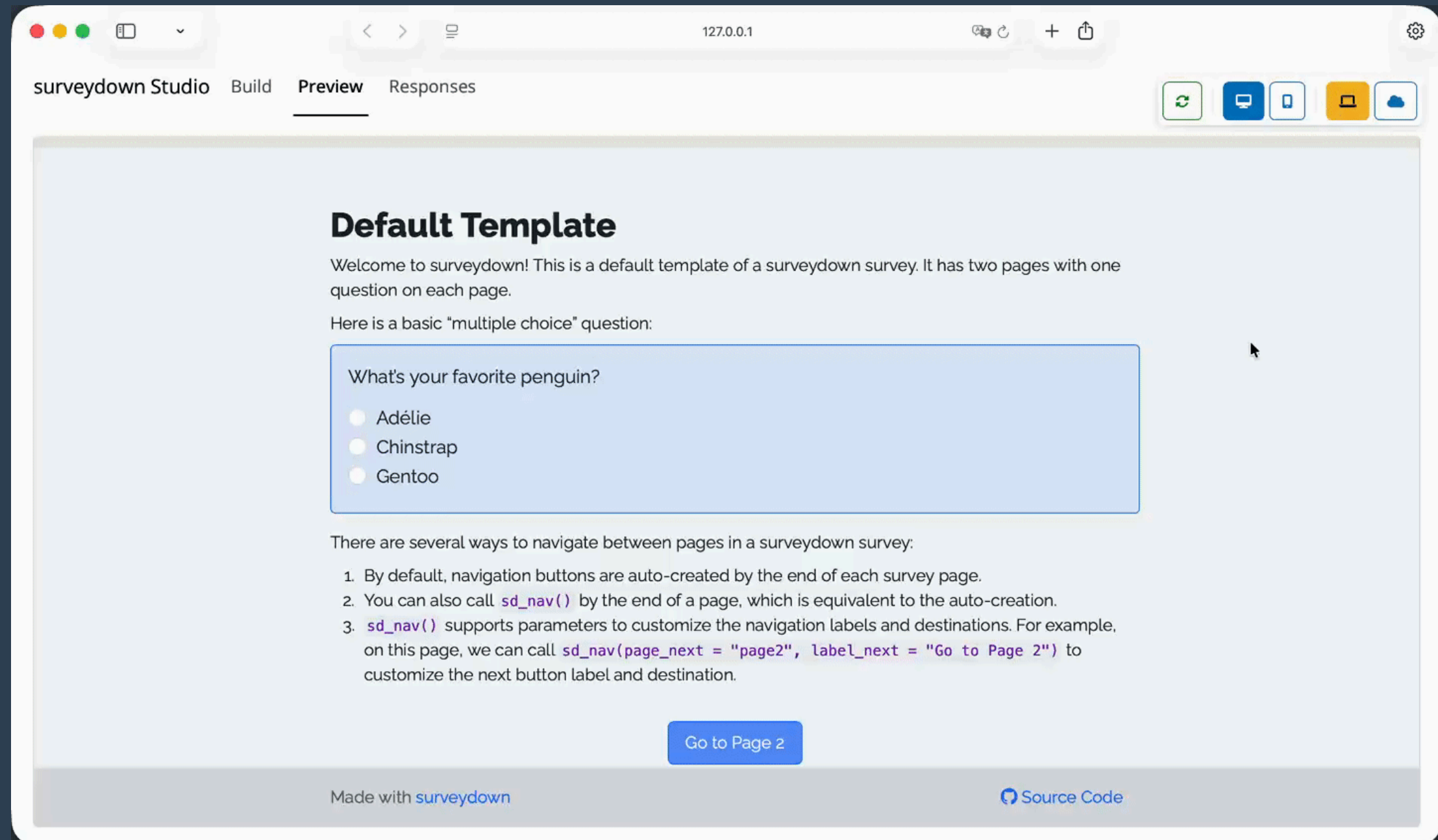
Code Panel: This panel shows the R code for the survey. It includes two files: 'survey.qmd' and 'app.R'. The 'survey.qmd' file contains the following code:

```
34- ---
35-
36-
37- --- page1
38-
39- # Title
40-
41- New texts.
42-
43- ```{r}
44- sd_question(
45-   type = 'mc',
46-   id   = 'penguins',
47-   label = "What's your favorite penguin?",
48-   option = c(
49-     'Adélie' = 'adelie',
50-     'Chinstrap' = 'chinstrap',
51-     'Gentoo' = 'gentoo'
52-   )
53- )
54- ---
55- --- page2
56-
57- This is page 2.
58-
59- ```{r}
60- sd_question(
61-   type = "text",
62-   id   = "text_question",
63-   label = "Write something:"
64- )
65- ---
```

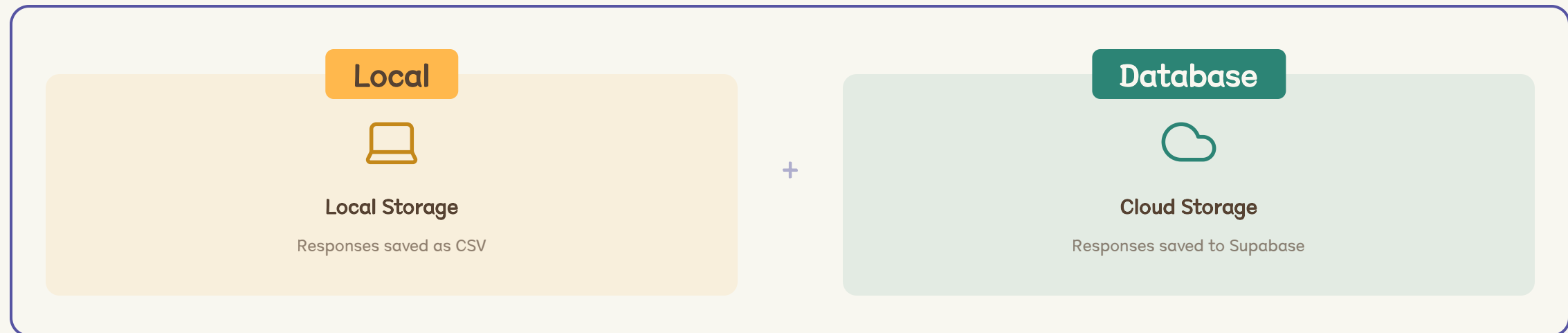
Views



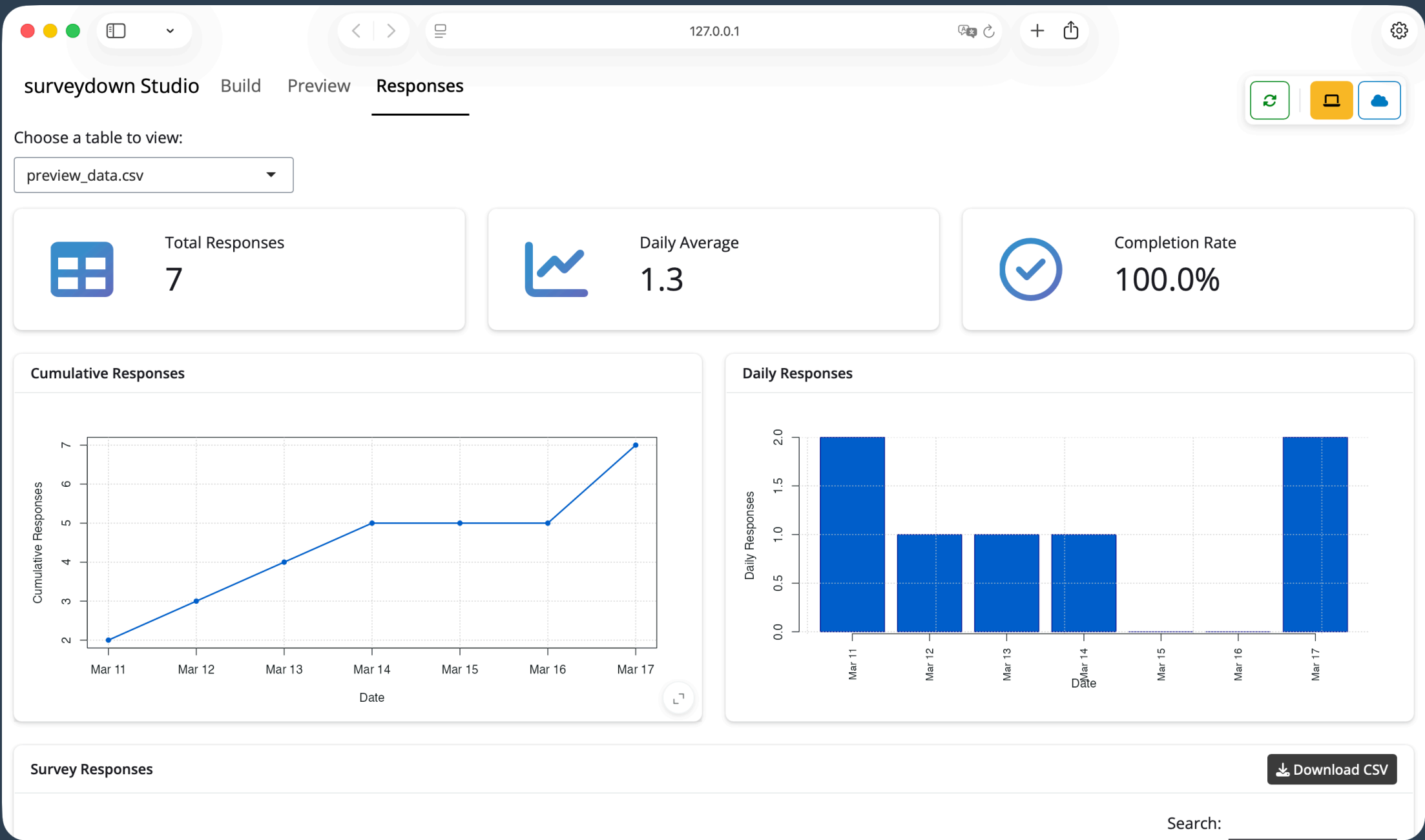
Desktop and Mobile Views



Responses



Local Responses



Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

127.0.0.1

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

127.0.0.1

Database Connection

Close

Not connected

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

default

Test Connection

Choose a table to view:

Select or type table name...

Database Responses

surveydown Studio

Build

Preview

Responses

127.0.0.1

Database Connection

Close

Connected (GSSAPI disabled)

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

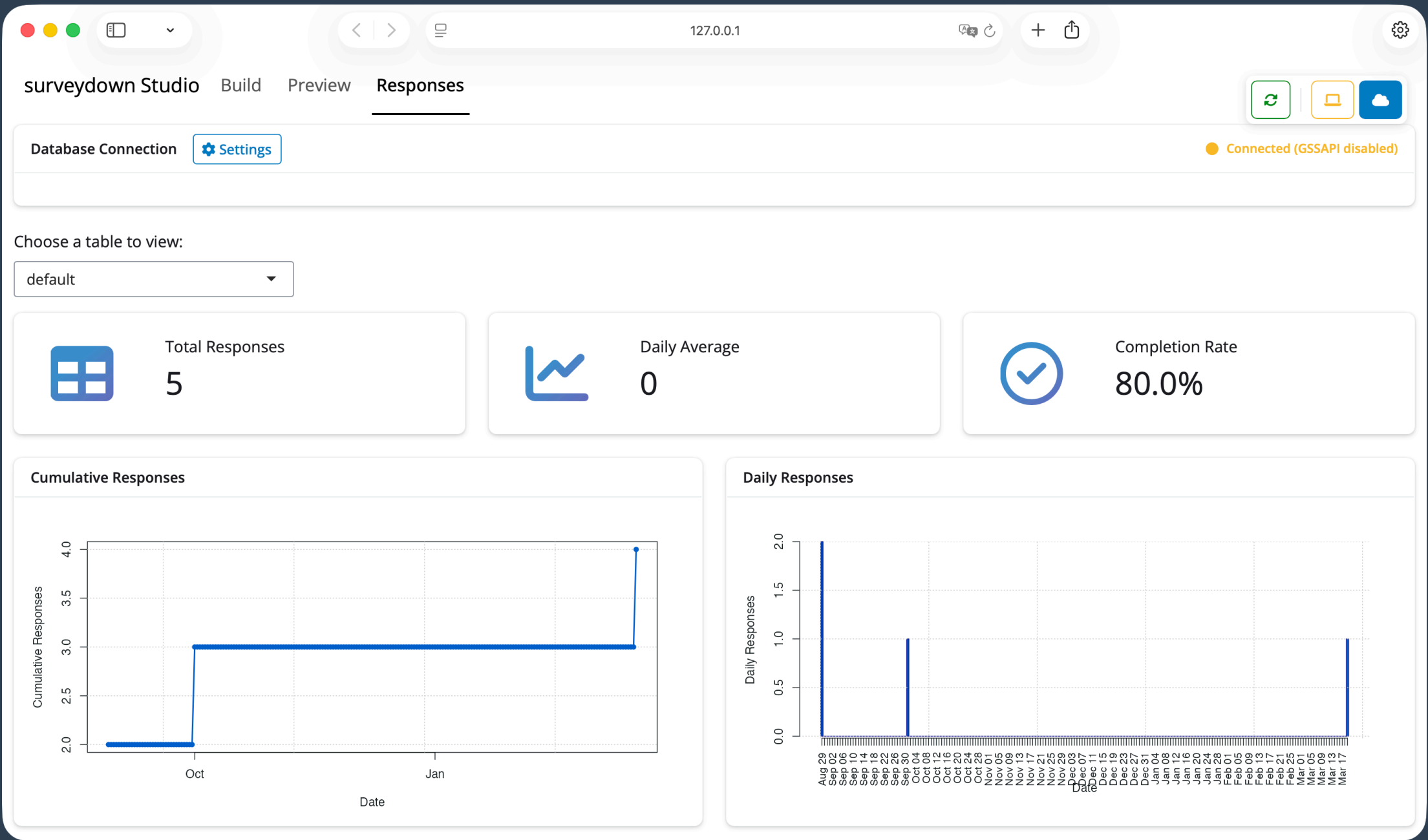
default

Test Connection

Choose a table to view:

default

Database Responses



sdstudio gives you...



GUI for Survey Editing



Live Preview



Response Dashboard



SQL Connection



Data View & Download



...and more!

surveydown + sdstudio

surveydown

 Text-based editing

 Published paper

 8000+ CRAN downloads

sdstudio

 GUI editing & live preview

 Actively developed

Open to contributions

 Better together!

Ways to help:

github.com/surveydown-dev/sdstudio/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute new features



Meet the team!



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea

surveydown.org

